

Binomial Identities

1. Consider the identity $(1+x)^4 = {}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4$.
 - a. By substituting $x = 1$, show that ${}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4$.
 - b. i. By substituting $x = -1$, show that ${}^4C_0 + {}^4C_2 + {}^4C_4 = {}^4C_1 + {}^4C_3$.
 ii. Hence, by using the result of part a., show that ${}^4C_0 + {}^4C_2 + {}^4C_4 = 2^3$.
 - c. i. Differentiate both sides of the identity.
 ii. By substituting $x = 1$, show that ${}^4C_1 + 2{}^4C_2 + 3{}^4C_3 + 4{}^4C_4 = 4 \times 2^3$.
 iii. By substituting $x = -1$, show that ${}^4C_1 - 2{}^4C_2 + 3{}^4C_3 - 4{}^4C_4 = 0$.
 - d. i. By integrating both sides of the identity, show that for some constant K , $\frac{1}{5}(1+x)^5 + K = {}^4C_0x + \frac{1}{2}{}^4C_1x^2 + \frac{1}{3}{}^4C_2x^3 + \frac{1}{4}{}^4C_3x^4 + \frac{1}{5}{}^4C_4x^5$.
 ii. By substituting $x = 0$, show that $K = -\frac{1}{5}$.
 iii. By substituting $x = -1$ in part i., show that ${}^4C_0 - \frac{1}{2}{}^4C_1 + \frac{1}{3}{}^4C_2 - \frac{1}{4}{}^4C_3 + \frac{1}{5}{}^4C_4 = \frac{1}{5}$.
2. Consider the identity $(1+x)^n = \sum_{k=0}^n {}^nC_k x^k$
 - a. By substituting $x = 1$, showing that $\sum_{k=0}^n {}^nC_k = 2^n$.
 - b. i. By substituting $x = -1$, show that ${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \dots$
 ii. Hence, by using the result of part a., show that ${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = 2^{n-1}$.
 - c. i. Differentiate both sides of the identity.
 ii. By substituting $x = 1$, show that $\sum_{k=1}^n k {}^nC_k = n2^{n-1}$.
 iii. By substituting $x = -1$, show that $\sum_{k=1}^n (-1)^{k-1} k {}^nC_k = 0$.
 - d. i. By integrating both sides of the identity, show that for some constant K , $\frac{1}{n+1}(1+x)^{n+1} + K = \sum_{k=0}^n {}^nC_k \frac{x^{k+1}}{k+1}$.
 ii. By substituting $x = 0$, show that $K = -\frac{1}{n+1}$.
 iii. By substituting $x = -1$ in part i., show that $\sum_{k=0}^n (-1)^k \frac{{}^nC_k}{k+2} = \frac{1}{n+1}$.
3. This question follows the same steps as question 2. Consider the identity $(1+x)^{2n} = \sum_{r=0}^{2n} {}^{2n}C_r x^r$.
 - a. Show that $\sum_{r=0}^{2n} {}^{2n}C_r = 2^{2n}$.
 - b. Show that ${}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_5 + \dots + {}^{2n}C_{2n-1} = 2^{2n-1}$.

Check both results on the Pascal triangle, using $n = 3$ and $n = 4$.

 - c. By differentiating both sides of the identity, show that:
 - i. $\sum_{r=1}^{2n} r {}^{2n}C_r = n2^{2n}$
 - ii. $\sum_{r=1}^{2n} (-1)^{r-1} r {}^{2n}C_r = 0$
 - d. By integrating both sides of the identity, show that:
 - i. $\sum_{r=0}^{2n} {}^{2n}C_r \frac{1}{r+1} = \frac{2^{2n+1}-1}{2n+1}$ [HINT: the constant of integration is $-\frac{1}{2n+1}$.]
 - ii. $\sum_{r=0}^{2n} (-1)^r \frac{{}^{2n}C_r}{r+1} = \frac{1}{2n+1}$.
4. a. By equating the coefficients of x^3 on the RHS and LHS of the identity $(1+x)^3(1+x)^9 = (1+x)^{12}$, show that ${}^3C_0 {}^9C_3 + {}^3C_1 {}^9C_2 + {}^3C_2 {}^9C_1 + {}^3C_3 {}^9C_0 = {}^{12}C_3$.
 b. By equating the coefficients of x^3 on the RHS and LHS of the identity $(1+x)^m(1+x)^n = (1+x)^{m+n}$, show that ${}^mC_0 {}^nC_3 + {}^mC_1 {}^nC_2 + {}^mC_2 {}^nC_1 + {}^mC_3 {}^nC_0 = {}^{m+n}C_3$.

5. a. Show that ${}^nC_k = {}^nC_{n-k}$
- b. By comparing coefficients of x^{10} on both sides of $(1+x)^{10}(1+x)^{10} = (1+x)^{20}$, show that $\sum_{k=0}^{10} ({}^{10}C_k)^2 = {}^{20}C_{10}$.
- c. By comparing coefficients of x^n on both sides of the identity $(1+x)^n(1+x)^n = (1+x)^{2n}$, show that $\sum_{k=0}^n ({}^nC_k)^2 = {}^{2n}C_n$.
- d. By comparing coefficients of x^{n+1} on both sides of $(1+x)^n(1+x)^n = (1+x)^{2n}$, show that ${}^nC_0 \times {}^nC_1 + {}^nC_1 \times {}^nC_2 + {}^nC_2 \times {}^nC_3 + \dots + {}^nC_{n-1} \times {}^nC_n = \frac{(2n)!}{(n-1)!(n+1)!}$.
- e. Prove that $(1+x)^n \left(1 + \frac{1}{x}\right)^n = \frac{1}{x^n} (1+x)^{2n}$. By equating coefficients of $\frac{1}{x}$, give an alternative proof of the result in part d.
6. a. By equating coefficients of $\frac{1}{x^4}$ in the expansion of $\left(1 + \frac{1}{x}\right)^4 \left(1 - \frac{1}{x}\right)^4 = \left(1 - \frac{1}{x^2}\right)^4$, prove that $({}^4C_0)^2 - ({}^4C_1)^2 + ({}^4C_2)^2 - ({}^4C_3)^2 + ({}^4C_4)^2 = {}^4C_2$.
- b. Generalise this result, and prove it, by considering the expansion of $\left(1 + \frac{1}{x}\right)^{2n} \left(1 - \frac{1}{x}\right)^{2n} = \left(1 - \frac{1}{x^2}\right)^{2n}$.
7. a. By expanding both side of the identity $(1+x)^{n+4} = (1+x)^n(1+x)^4$, show that $\binom{n+4}{r} = \binom{n}{r} + 4\binom{n}{r-1} + 6\binom{n}{r-2} + 4\binom{n}{r-3} + \binom{n}{r-4}$, and state the necessary restriction on r .
- b. By expanding both sides of the identity $(1+x)^{p+q} = (1+x)^p(1+x)^q$, show that $\binom{p+q}{r} = \binom{p}{r} + \binom{p}{r-1}\binom{q}{1} + \binom{p}{r-2}\binom{q}{2} + \dots + \binom{p}{1}\binom{q}{r-1} + \binom{q}{r}$, and state the necessary restriction on r .
8. a. By considering the expansion of $(1+x)^n$, show that:
- $\sum_{k=0}^n {}^nC_k = 2^n$
 - $\sum_{k=1}^n k {}^nC_k = n2^{n-1}$
- b. Hence show that $\sum_{k=0}^n (k+1) {}^nC_k = 2^{n-1}(n+2)$.
9. a. Find the coefficient of x^{n+r} in the expansion of $(1+x)^{3n}$.
- b. By writing $(1+x)^{3n}$ as $(1+x)^n(1+x)^{2n}$, prove that for $0 < r \leq n$, $\binom{n}{0}\binom{2n}{r} + \binom{n}{1}\binom{2n}{r+1} + \binom{n}{2}\binom{2n}{r+2} + \dots + \binom{n}{n}\binom{2n}{r+n} = \binom{3n}{n+r}$.
10. a. Evaluate $\int_0^4 (1+x)^n dx$
- b. By expanding $(1+x)^n$ and then integrating, show that $\sum_{r=0}^n \frac{4^{r+1}}{r+1} {}^nC_r = \frac{5^{n+1}-1}{n+1}$.
11. a. Show that $x^n(1+x)^n \left(1 + \frac{1}{x}\right)^n = (1+x)^{2n}$
- b. Hence prove that $1 + \binom{n}{1}^2 + \binom{n}{2}^2 + \dots + \binom{n}{n}^2 = \binom{2n}{n}$.
12. If $(1+x)^n = c_0 + c_1x + c_2x^2 + \dots + c_nx^n$, show that $c_0c_2 + c_1c_3 + c_2c_4 + \dots + c_{n-2}c_n = \frac{(2n)!}{(n+2)!(n-2)!}$.

Answer

1. c. i. $4(1+x)^3 = {}^4C_1 + 2{}^4C_2x + 3{}^4C_3x^2 + 4{}^4C_4x^3$
6. b. $({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + ({}^{2n}C_{2n})^2 = (-1)^n {}^{2n}C_n$
7. a. $4 \leq r \leq n$; b. $r \leq p$ and $r \leq q$
9. a. ${}^{3n}C_{n+r}$
10. a. $\frac{5^{n+1}-1}{n+1}$